

SAT Math

Circles 2

Question # ID

Answer

2.1 8e7689e0

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

4

2.2 74d8b897

An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

81

2.3 856372ca

In the xy -plane, a circle with radius 5 has center $(-8, 6)$. Which of the following is an equation of the circle?

A. $(x - 8)^2 + (y + 6)^2 = 25$

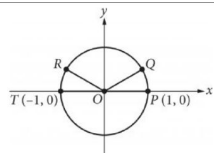
B. $(x + 8)^2 + (y - 6)^2 = 25$

C. $(x - 8)^2 + (y + 6)^2 = 5$

D. $(x + 8)^2 + (y - 6)^2 = 5$

B

2.4 95ba2d09



In the xy -plane above, points P , Q , R , and T lie on the circle with center O . The degree measures of angles POQ and ROT are each 30° . What is the radian measure of angle QOR ?

A. $\frac{5}{6}\pi$

B. $\frac{3}{4}\pi$

C. $\frac{2}{3}\pi$

D. $\frac{1}{3}\pi$

C

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Question #	ID		Answer
2.5	82c8325f	A circle in the xy-plane has its center at $(-4, 5)$ and the point $(-8, 8)$ lies on the circle. Which equation represents this circle? A. $(x - 4)^2 + (y + 5)^2 = 5$ B. $(x + 4)^2 + (y - 5)^2 = 5$ C. $(x - 4)^2 + (y + 5)^2 = 25$ D. $(x + 4)^2 + (y - 5)^2 = 25$	D
2.6	a0cacec1	An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in <u>degrees</u>?	192
2.7	b96ff36e	In the xy-plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. The point $(6, c)$, where c is a constant, lies on this circle. What is the value of c?	5
2.8	2d521ca9	The measure of angle Z is 60°. What is the measure, in <u>radians</u>, of angle Z? A. $\frac{1}{6}\pi$ B. $\frac{1}{3}\pi$ C. $\frac{2}{3}\pi$ D. 1π	B
2.9	ee540927	$x^2 + 58x + y^2 = 0$ In the xy-plane, the graph of the given equation is a circle. What are the coordinates (x, y) of the center of the circle? A. $(0, 29)$ B. $(0, -29)$ C. $(29, 0)$ D. $(-29, 0)$	D

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Question # ID
2.10 fc8aa563

Answer

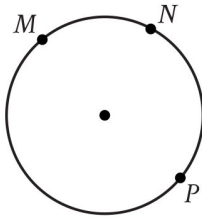
What is the center of the circle in the xy -plane defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$?

- A. $(-1, -7)$
- B. $(-1, 7)$
- C. $(1, -7)$
- D. $(1, 7)$

C

2.11 800e71b8

D



Points M , N , and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

- A. 39
- B. 156
- C. 195
- D. 234